

American Samoa Model Checks

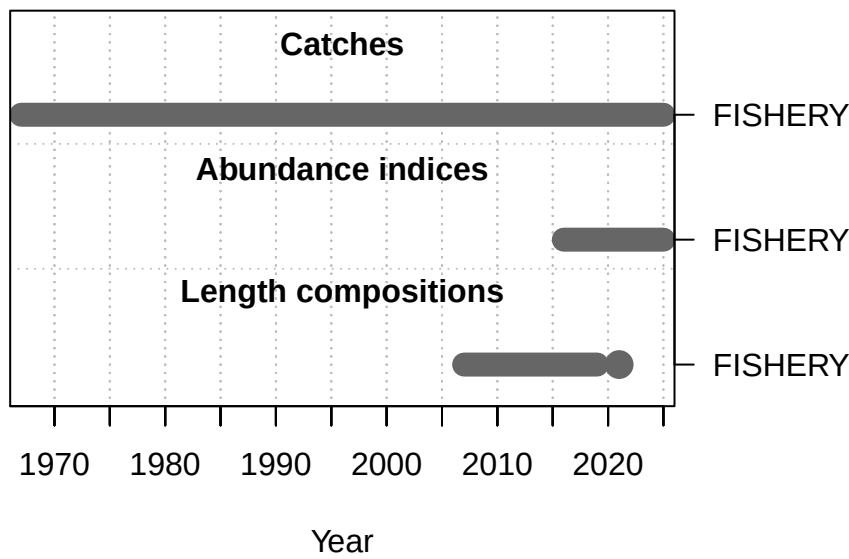
Marc Nadon and Meg Oshima

2026-04-06

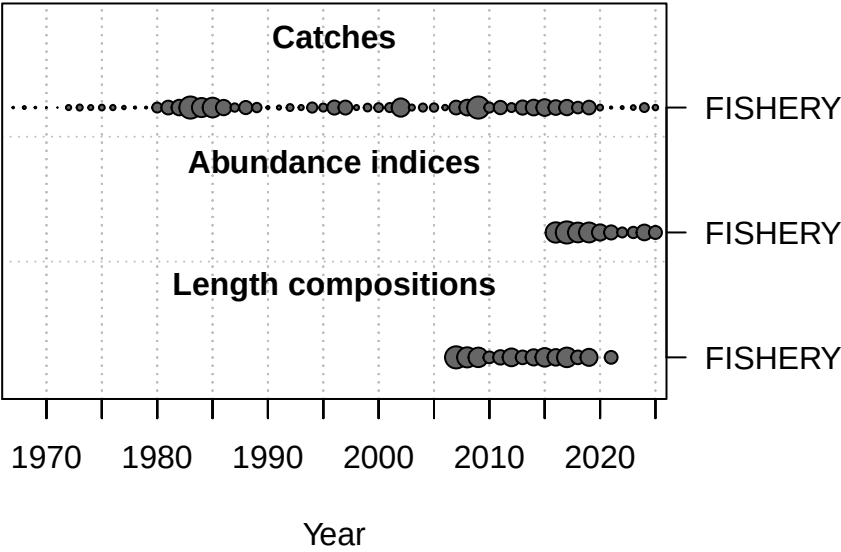
This is a summary report for the APRU base model run.

Model Output

Input Data



Fleet	RMSE.perc	Nobs
FISHERY	91.5	10
Combined	91.5	10



Convergence Check

```

Converged      MaxGrad
1      TRUE 1.43385e-05
```

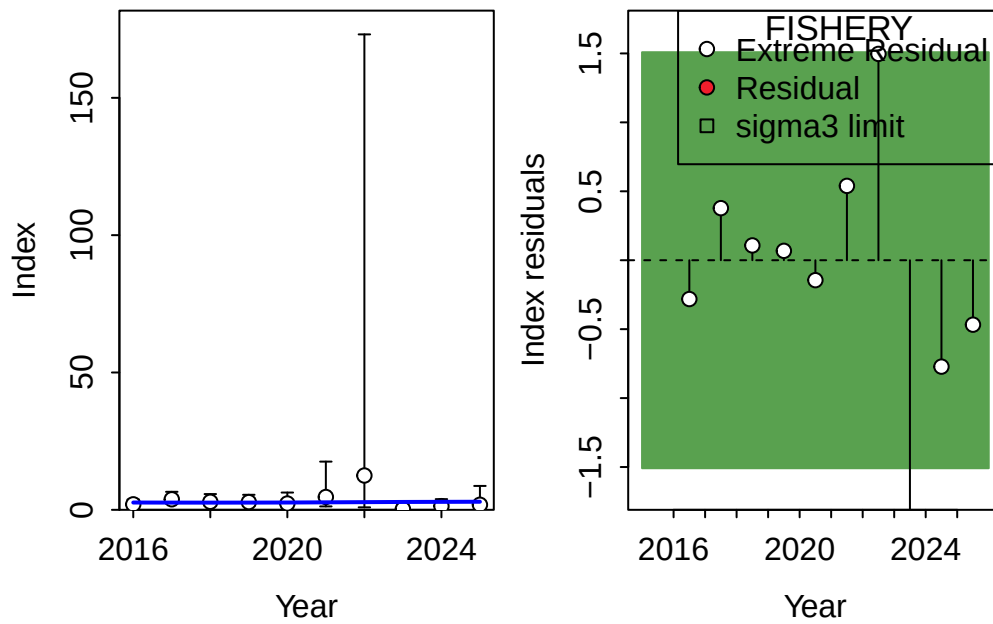
```

[1] "1 NOTE:  Max data length bin: 90  < max pop len bins: 100; so will accumulate larger pop
[2] "N warnings: 1"
```

Fleet	RMSE.perc	Nobs
FISHERY	4.8	14
Combined	4.8	14

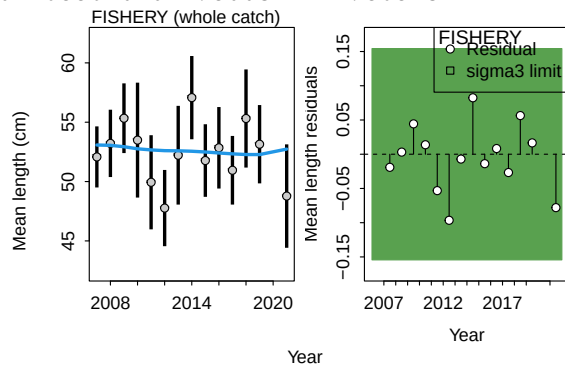
Fit to Model

CPUE

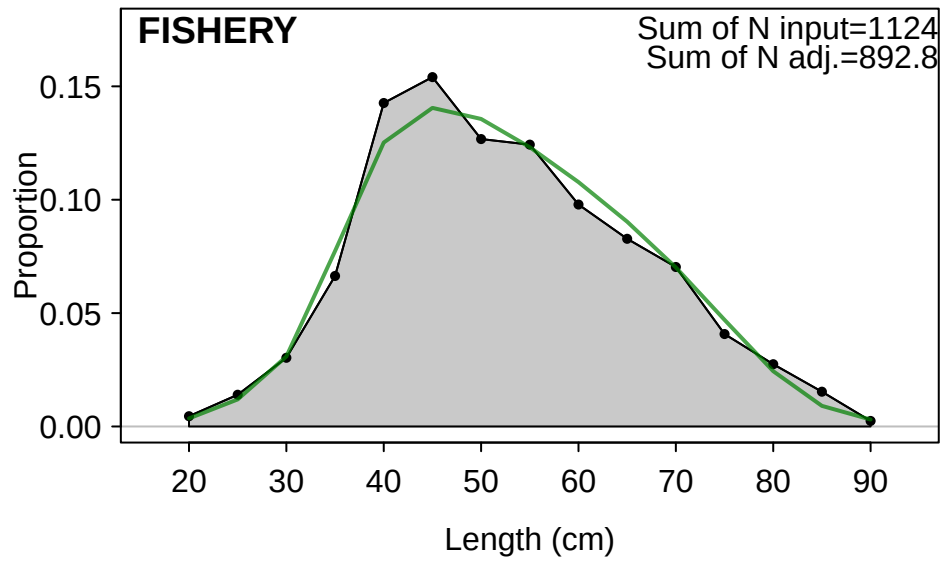


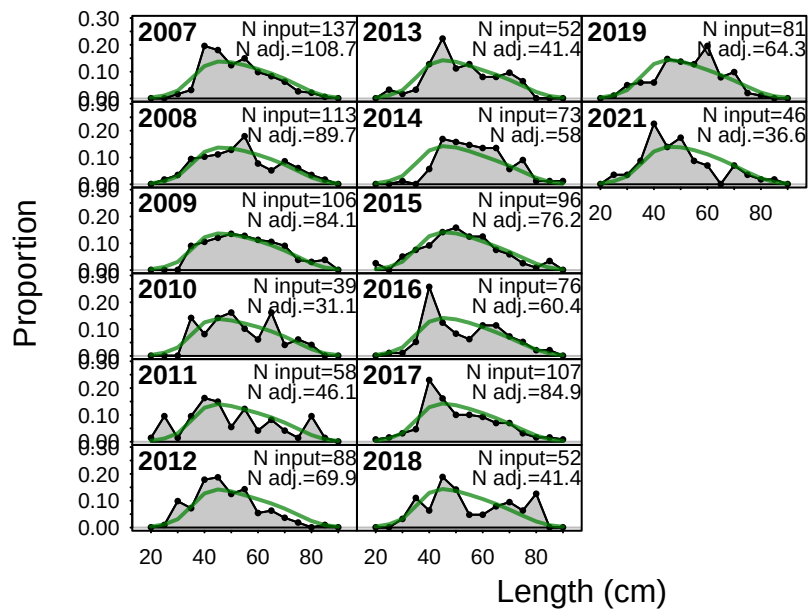
w lo hi
0.4685640 0.2750032 1.2786543

Index runs.p test sigma3.lo sigma3.hi type
1 FISHERY 0.711 Passed -0.1543101 0.1543101 len



Length Comp

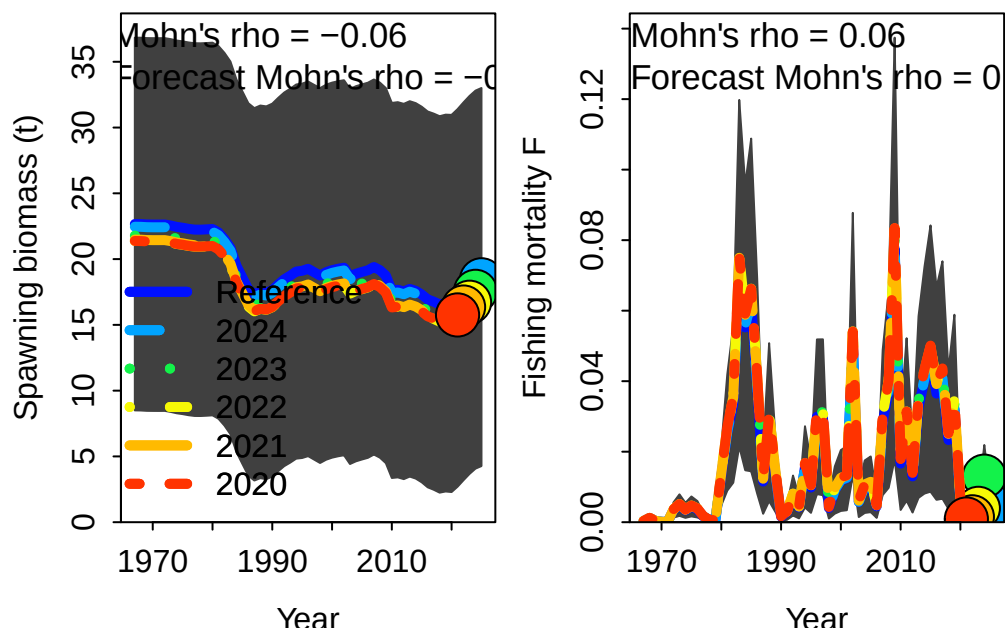




Retrospective and Hindcasting

Retrospective

Mohn's Rho stats, including one step ahead forecasts:



Mohn's Rho stats, including one step ahead forecasts:

	type	peel	Rho	ForecastRho
1	F	2024	0.01128672	0.01115067
2	F	2023	0.05027253	0.04954419
3	F	2022	0.07783612	0.07606512
4	F	2021	0.07252286	0.07061850
5	F	2020	0.08060166	0.07848917
6	F Combined		0.05850398	0.05717353

Hindcasting

Plotting Hindcast Cross-Validation (one-step-ahead)

Computing MASE with all 5 of 5 prediction residuals for Index FISHERY

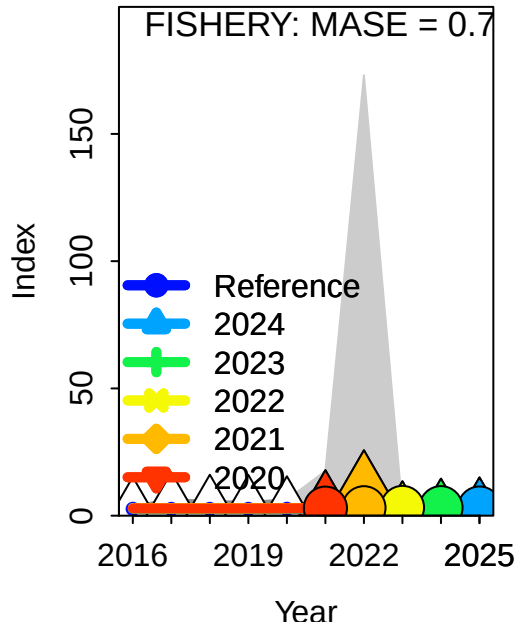
MASE stats by Index:

Plotting Hindcast Cross-Validation (one-step-ahead)

No observations in evaluation years to compute prediction residuals for Index FISHERY

MASE stats by Index:

	Index	Season	MASE	MAE.PR	MAE.base	MASE.adj	n.eval
1	FISHERY	1	NA	NA	NA	NA	0



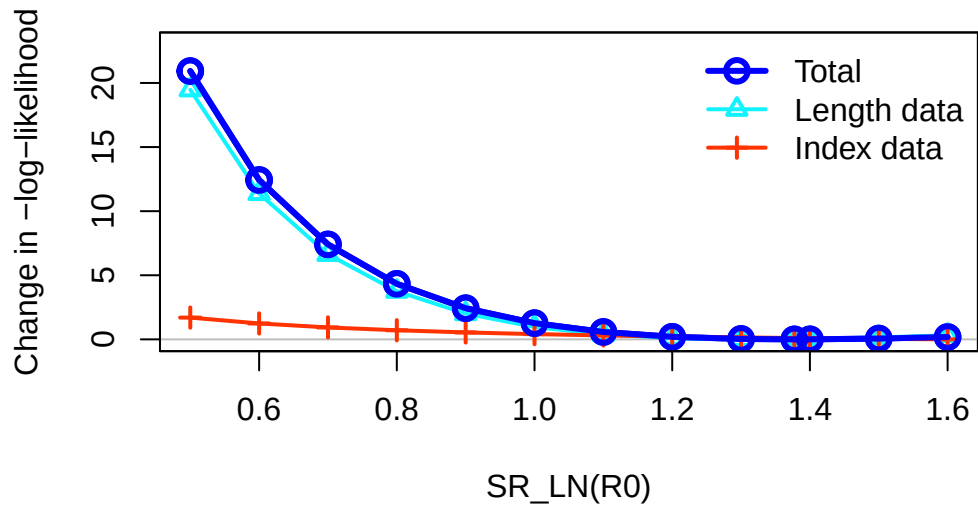
Recruitment Deviations

Likelihood Profile

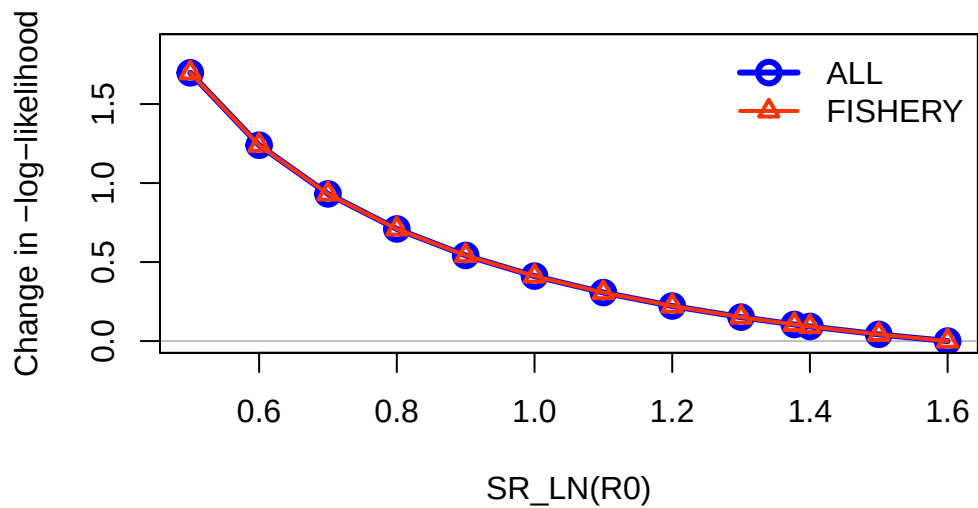
[1] "SR_LN(R0)"

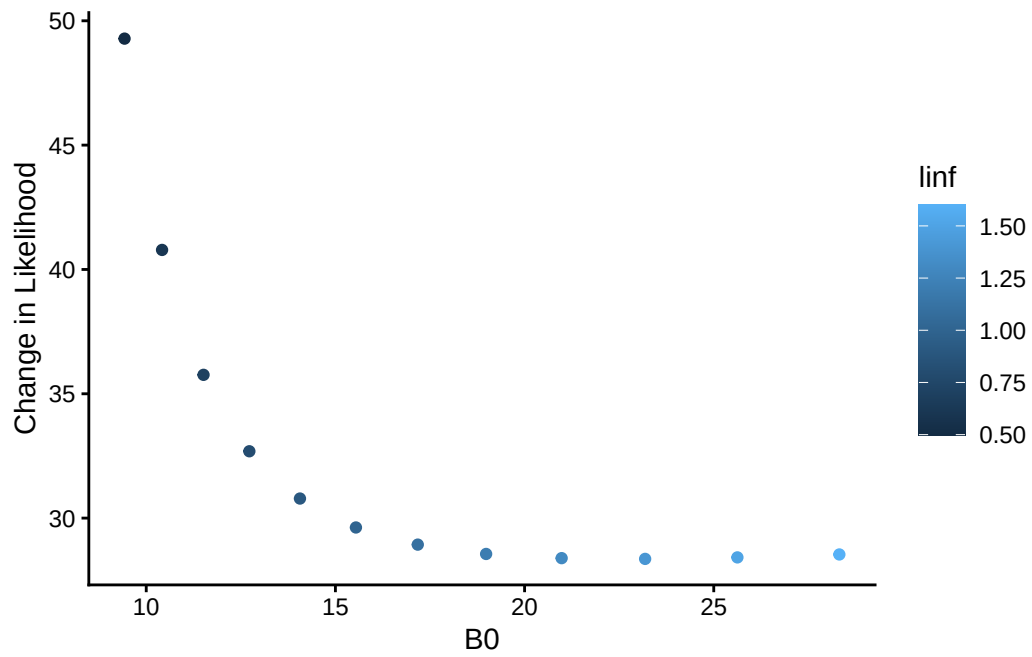
	frac_change	include	label
TOTAL	1.0000	TRUE	Total
Catch	0.0000	FALSE	Catch
Equil_catch	0.0000	FALSE	Equilibrium catch
Survey	0.0811	TRUE	Index data
Length_comp	0.9303	TRUE	Length data
Recruitment	0.0000	FALSE	Recruitment
InitEQ_Regime	0.0000	FALSE	Initital equilibrium recruitment
Forecast_Recruitment	0.0000	FALSE	Forecast recruitment

Parm_priors	0.0060	FALSE	Priors
Parm_softbounds	0.0000	FALSE	Soft bounds
Parm_devs	0.0000	FALSE	Parameter deviations
Crash_Pen	0.0000	FALSE	Crash penalty



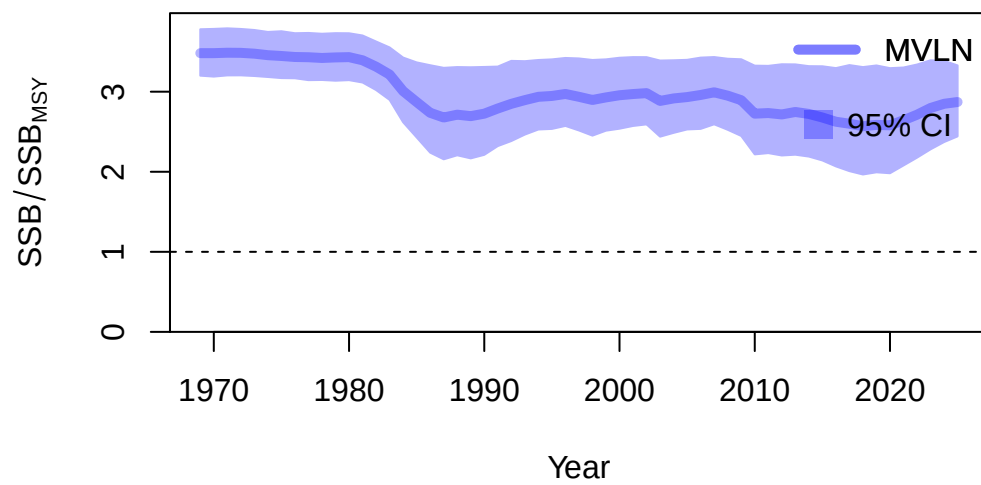
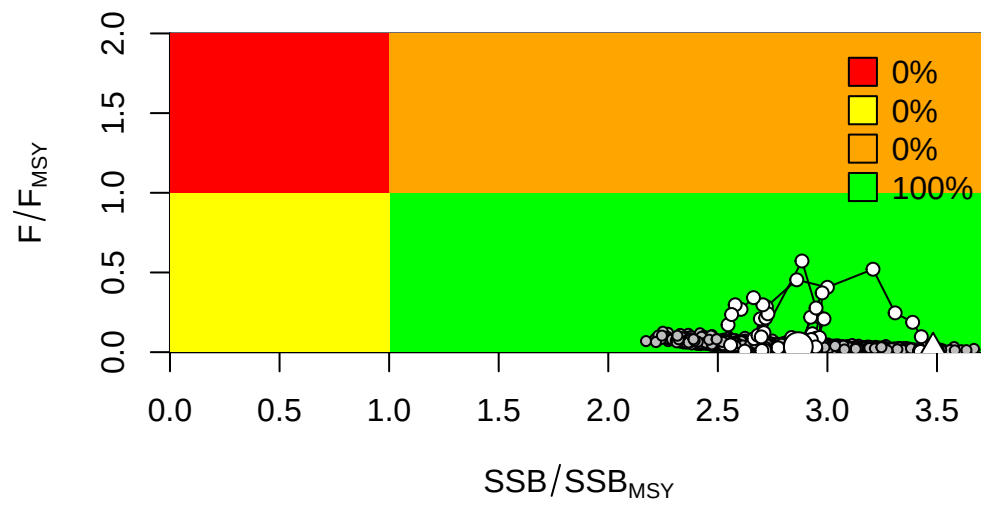
Changes in survey likelihood by fleet

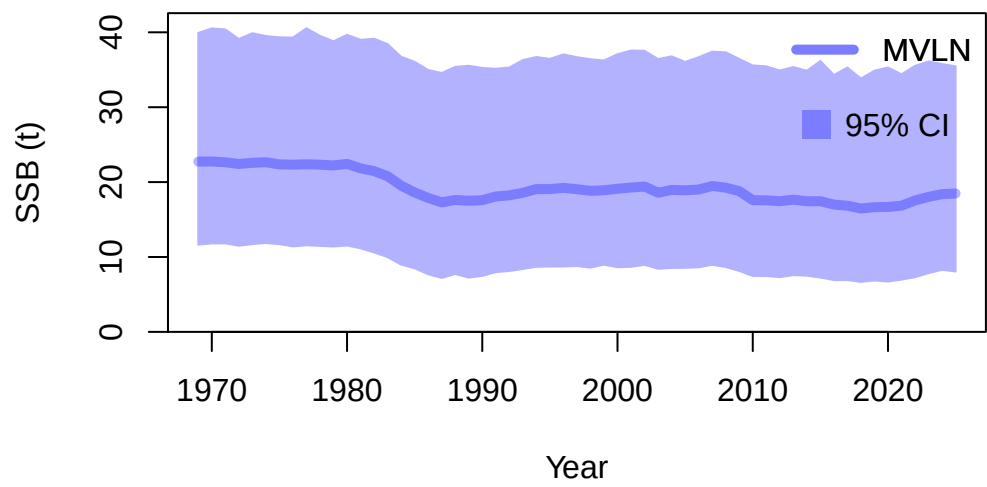
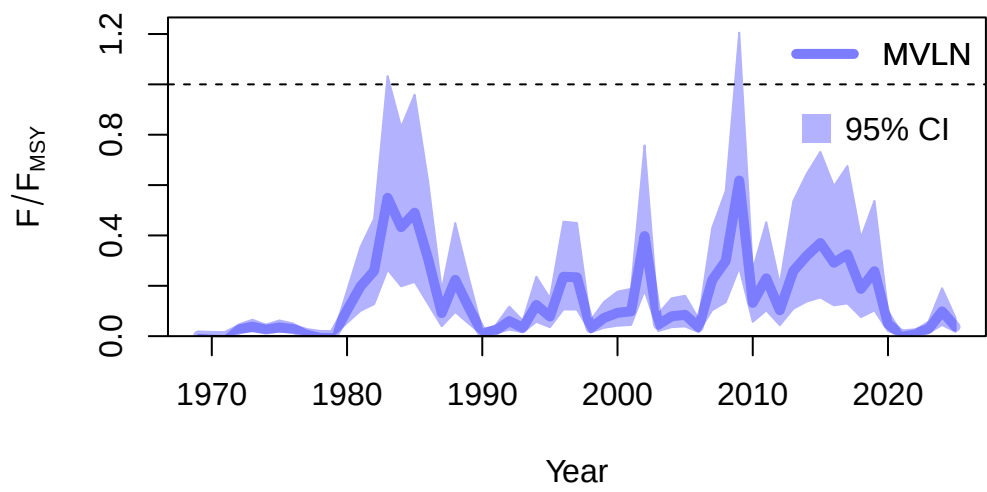


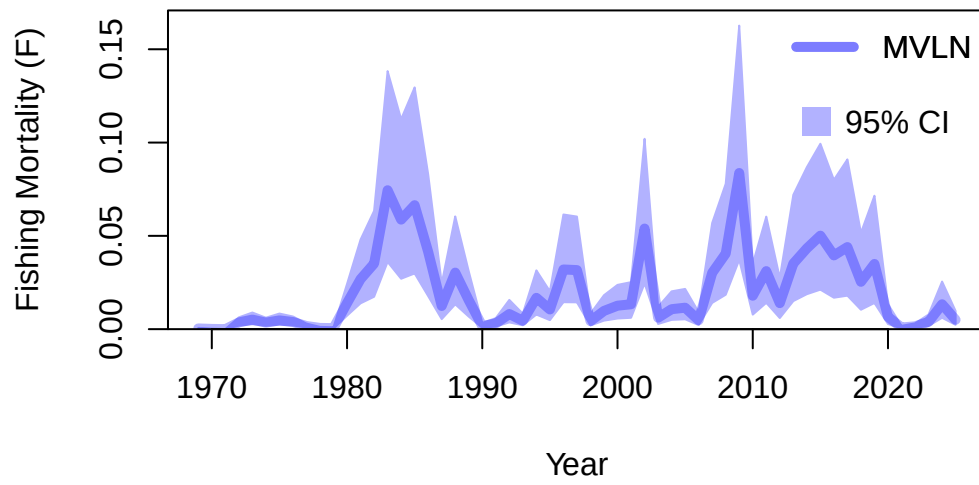


Management Quantities

starter.sso with Bratio: SSB/SSB_{MSY} and F: $_{abs_F}$

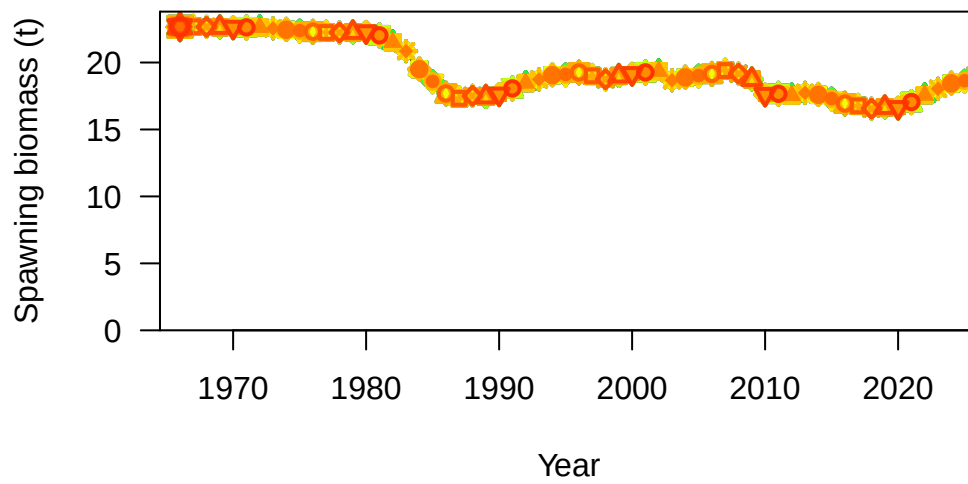
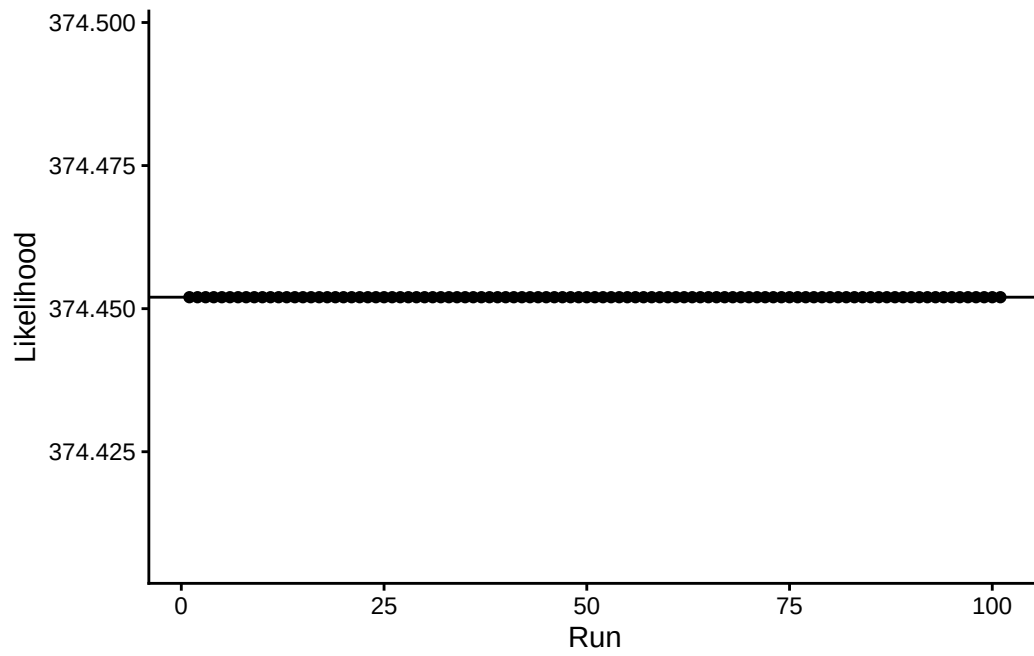


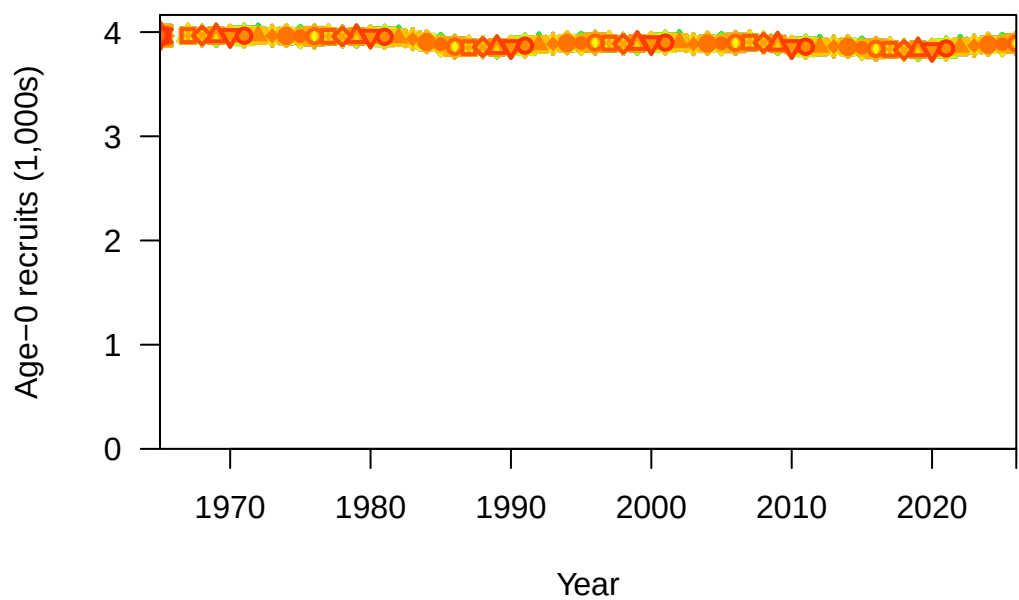
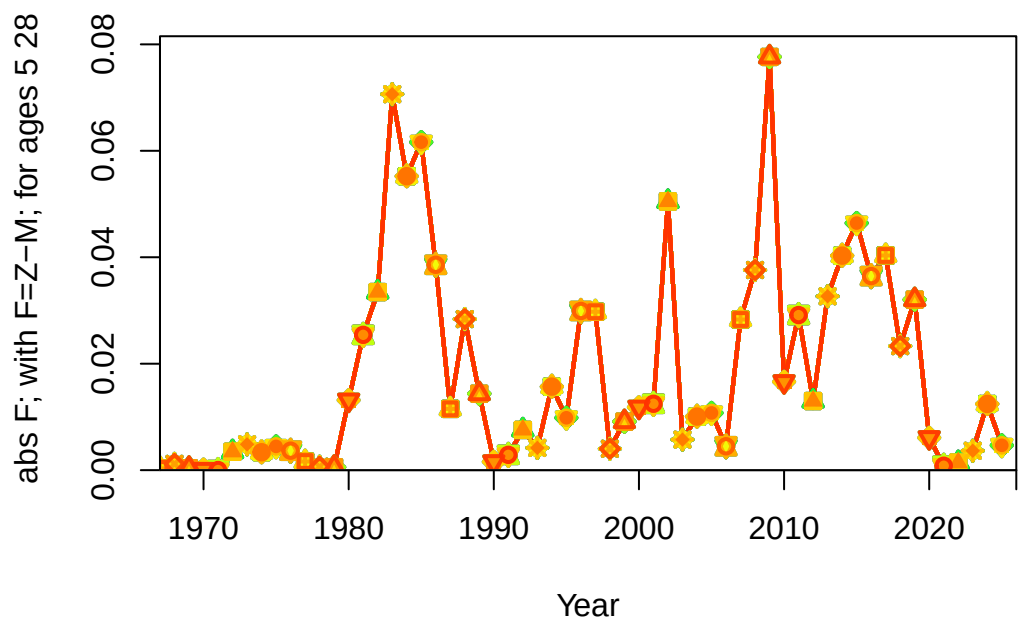




null device
1

Jitter





Selectivity and Maturity

